# 08-02 第1篇

**The Social Structure of Ants**

Ants are among the most successful insects on Earth, living in colonies that can contain millions of members. Every ant colony has a clear social structure with different roles for different ants. At the top of this structure is the queen, whose main job is to lay eggs and ensure the colony's survival. Worker ants, which are all female, do most of the daily work such as gathering food, caring for the young, and defending the nest. [1] Male ants appear only during mating season and typically live for a short time. This division of labor allows ant colonies to work efficiently and survive in nearly every environment on our planet.

The queen ant is the heart of the colony. She can live for many years—some species have queens that live over 20 years. During her lifetime, a single queen can produce millions of eggs. [2] Without the queen, the colony would quickly fall apart because no new workers could be born. Interestingly, the queen decides whether an egg will become a worker or a new queen by controlling the food she gives to the larvae.

Worker ants have different roles based on their age and size. Young workers usually stay inside the nest to care for the queen and baby ants. [3] They feed the larvae, clean the nest, and remove waste. Older and larger workers often become foragers, going outside to search for food. Some species even have soldier ants with larger heads and stronger jaws, specialized for defending the colony from enemies.

Communication is key to ant society. Ants use chemicals called pheromones to send messages to each other. When a worker finds food, she leaves a scent trail so others can follow. [4] Ants also touch their antennae to share information. This efficient communication system helps the entire colony work as one organized unit, almost like a superorganism.

[5] Understanding how ants live and work together can teach us about teamwork and organization. Their societies show us that even small creatures can achieve great things when every member contributes to the common goal.

## 选项

A. They mate with new queens and then die soon after, having served their single purpose. B. The colony depends entirely on her egg-laying ability for its continued existence. C. As they grow, their duties change and they move to different tasks within the colony. D. Different pheromones can mean different things—alarm, food location, or trail marking. E. Scientists study ants to learn more about cooperation and social

behavior in nature. F. For example, leaf-cutter ants in South America farm fungus to feed their colonies.

## 答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

## 深度解析

- **[1] A**：
  - **线索**：空前提及 worker ants 的角色，空后无内容，本段末尾。本段在介绍蚁群社会结构：queen、worker ants，接着应介绍 male ants。选项 A 说明雄蚁只在交配季节出现、寿命短、完成交配任务后死亡，与段落对三类蚂蚁的逐一介绍完全吻合。
- **[2] B**：
  - **线索**：空前描述 queen 产卵能力，空后说“Without the queen, the colony would quickly fall apart”，前后需因果衔接。选项 B 强调“colony depends entirely on her egg-laying ability”，直接对应“no new workers could be born”，形成逻辑闭环。
- **[3] C**：
  - **线索**：空前说 young workers 照顾 queen 和 baby ants，空后列举“They feed the larvae, clean the nest...”，需承上启下的过渡。选项 C 说明“As they grow, their duties change”，既承接年龄/角色变化，又自然引出后文的 foraging 和 soldier ants，符合段落结构。
- **[4] D**：
  - **线索**：空前列举 pheromone 用途（如 scent trail），空后提到“touch their antennae to share information”，需补充 pheromone 的更多功能。选项 D 列举“alarm, food location, or trail marking”，与前文 scent trail 呼应，扩展说明化学信号的多样性。
- **[5] E**：
  - **线索**：结尾段需总结全文、升华主题。全文讲蚂蚁的社会结构与分工，选项 E 提出“Scientists study ants to learn about cooperation and social behavior”，符合说明文收尾方式，且“cooperation”与“teamwork and organization”语义一致。
- **干扰项分析**：

- **F. For example, leaf-cutter ants in South America farm fungus to feed their colonies.**：虽与蚂蚁主题相关，但属于具体物种举例，与前文对不同角色（queen, workers, males）的总体介绍层次不符；且插入任一空位都会打断段落的逻辑推进，易在 [3] 或 [4] 处被误选。